

Mathematical Foundation of Computer Science

Detailed Solutions for Selected Questions (Total: 56 Marks)

Question 1

14 Marks

(a) Equivalence Relation

[7 Marks]

Problem: Let $X = \{1, 2, 3, 4, 5, 6, 7\}$ and $R = \{(x, y) \mid x - y \text{ is divisible by } 3\}$. Show that R is an equivalence relation.

To show that R is an equivalence relation, we must prove it is reflexive, symmetric, and transitive.

1. Reflexivity:

For any $x \in X$, consider $x - x = 0$.

Since $0 = 0 \times 3$, 0 is divisible by 3.

Therefore, $(x, x) \in R$ for all $x \in X$. Hence, R is **reflexive**.

2. Symmetry:

Let $(x, y) \in R$. This means $x - y$ is divisible by 3.

So, $x - y = 3k$ for some integer k .

Multiplying by -1 , we get $-(x - y) = -3k \implies y - x = 3(-k)$.

Since $-k$ is an integer, $y - x$ is divisible by 3.

Therefore, $(y, x) \in R$. Hence, R is **symmetric**.

3. Transitivity:

Let $(x, y) \in R$ and $(y, z) \in R$.

This implies $x - y = 3k$ (for some integer k) and $y - z = 3m$ (for some integer m).

Adding the two equations: $(x - y) + (y - z) = 3k + 3m$

$x - z = 3(k + m)$

Since $(k + m)$ is an integer, $x - z$ is divisible by 3.

Therefore, $(x, z) \in R$. Hence, R is **transitive**.

Conclusion: Since R is reflexive, symmetric, and transitive, it is an equivalence relation.

(b) Transitive Closure and Digraph**[7 Marks]**

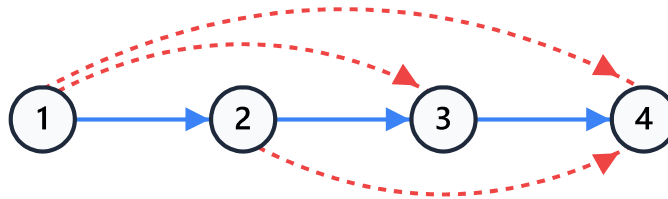
Problem: Let $A = \{1, 2, 3, 4\}$ and $R = \{(1, 2), (2, 3), (3, 4)\}$ be a relation on A . Find the transitive closure of R and draw its digraph.

Finding the Transitive Closure (R^* or R^+):

The relation is given as $R = \{(1, 2), (2, 3), (3, 4)\}$.

- Since $(1, 2) \in R$ and $(2, 3) \in R$, we must add $(1, 3)$.
- Since $(2, 3) \in R$ and $(3, 4) \in R$, we must add $(2, 4)$.
- Now we have $(1, 3)$ and $(3, 4) \in R$, we must add $(1, 4)$.

Transitive Closure $R^* = \{(1, 2), (2, 3), (3, 4), (1, 3), (2, 4), (1, 4)\}$

Digraph of Transitive Closure:

Note: Solid blue lines represent original relation R . Dashed red lines represent added edges for transitive closure.

Question 2**14 Marks****(a) Functions and their Properties****[7 Marks]**

Problem: List all possible functions from the set $X = \{a, b, c\}$ to the set $Y = \{0, 1\}$. Indicate in each case whether the function is one-one, onto, or both.

Total number of functions from X to Y is $|Y|^{|X|} = 2^3 = 8$.

FUNCTION NAME	MAPPING $\{(X, F(X))\}$	ONE-ONE (INJECTIVE)?	ONTO (SURJECTIVE)?	BOTH (BIJECTIVE)?
f_1	$\{(a, 0), (b, 0), (c, 0)\}$	No	No (only maps to 0)	No
f_2	$\{(a, 0), (b, 0), (c, 1)\}$	No	Yes	No
f_3	$\{(a, 0), (b, 1), (c, 0)\}$	No	Yes	No
f_4	$\{(a, 0), (b, 1), (c, 1)\}$	No	Yes	No
f_5	$\{(a, 1), (b, 0), (c, 0)\}$	No	Yes	No
f_6	$\{(a, 1), (b, 0), (c, 1)\}$	No	Yes	No
f_7	$\{(a, 1), (b, 1), (c, 0)\}$	No	Yes	No
f_8	$\{(a, 1), (b, 1), (c, 1)\}$	No	No (only maps to 1)	No

Reasoning: None of the functions are one-one because the domain size (3) is strictly greater than the codomain size (2). By the Pigeonhole Principle, at least two elements in X must map to the same element in Y .

(b) Characteristic Functions

[7 Marks]

Problem: Prove that: (i) $f_{A \cap B} = f_A \cdot f_B$ and (ii) $f_{A \cup B} = f_A + f_B - f_A f_B$
When f is a characteristic function.

The characteristic function of a set $A \subseteq U$ is defined as:

$$f_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

Proof of (i) $f_{A \cap B} = f_A \cdot f_B$:

We evaluate both sides for an arbitrary element $x \in U$ in two cases:

- **Case 1 ($x \in A \cap B$):** This means $x \in A$ and $x \in B$.

$$\text{LHS: } f_{A \cap B}(x) = 1$$

$$\text{RHS: } f_A(x) \cdot f_B(x) = 1 \cdot 1 = 1. \text{ Both sides are equal.}$$

- **Case 2 ($x \notin A \cap B$):** This means $x \notin A$ or $x \notin B$.

$$\text{LHS: } f_{A \cap B}(x) = 0$$

$$\text{RHS: Either } f_A(x) = 0 \text{ or } f_B(x) = 0, \text{ so their product } f_A(x) \cdot f_B(x) = 0. \text{ Both sides are equal.}$$

Hence, $f_{A \cap B} = f_A \cdot f_B$ is proved.

Proof of (ii) $f_{A \cup B} = f_A + f_B - f_A f_B$:

We consider four possible cases for an element x regarding its membership in sets A and B:

CONDITION FOR X	$F_A(X)$	$F_B(X)$	LHS: $F_{A \cup B}(X)$	RHS: $F_A(X) + F_B(X) - F_A(X)F_B(X)$
$x \in A, x \in B$	1	1	1	$1 + 1 - (1)(1) = 2 - 1 = \mathbf{1}$
$x \in A, x \notin B$	1	0	1	$1 + 0 - (1)(0) = 1 - 0 = \mathbf{1}$
$x \notin A, x \in B$	0	1	1	$0 + 1 - (0)(1) = 1 - 0 = \mathbf{1}$
$x \notin A, x \notin B$	0	0	0	$0 + 0 - (0)(0) = 0 - 0 = \mathbf{0}$

Since LHS = RHS for all possible cases, $f_{A \cup B} = f_A + f_B - f_A f_B$ is proved.

Question 3

14 Marks

(a) Fuzzy Sets Operations

[7 Marks]

Problem: If $U = \{a, b, c, d\}$, A and B are two fuzzy sets:

$$A = \{(a, 0.5), (b, 0.8), (c, 0.0), (d, 0.3)\}$$

$$B = \{(a, 0.2), (b, 1.0), (c, 0.1), (d, 0.7)\}$$

Find (i) $A \cup B$, (ii) $A \cap B$ and show that $(A \cup B)' = A' \cap B'$.

(i) Finding $A \cup B$:

Fuzzy union uses the max operator: $(A \cup B)(x) = \max[A(x), B(x)]$.

- $x = a: \max(0.5, 0.2) = 0.5$
- $x = b: \max(0.8, 1.0) = 1.0$
- $x = c: \max(0.0, 0.1) = 0.1$
- $x = d: \max(0.3, 0.7) = 0.7$

$$A \cup B = \{(a, 0.5), (b, 1.0), (c, 0.1), (d, 0.7)\}$$

(ii) Finding $A \cap B$:

Fuzzy intersection uses the min operator: $(A \cap B)(x) = \min[A(x), B(x)]$.

- $x = a: \min(0.5, 0.2) = 0.2$
- $x = b: \min(0.8, 1.0) = 0.8$
- $x = c: \min(0.0, 0.1) = 0.0$
- $x = d: \min(0.3, 0.7) = 0.3$

$$A \cap B = \{(a, 0.2), (b, 0.8), (c, 0.0), (d, 0.3)\}$$

Showing that $(A \cup B)' = A' \cap B'$ (De Morgan's Law):

First, calculate LHS: $(A \cup B)'(x) = 1 - (A \cup B)(x)$. Using our result from (i):

- $x = a: 1 - 0.5 = 0.5$
- $x = b: 1 - 1.0 = 0.0$
- $x = c: 1 - 0.1 = 0.9$
- $x = d: 1 - 0.7 = 0.3$

$$\text{LHS: } (A \cup B)' = \{(a, 0.5), (b, 0.0), (c, 0.9), (d, 0.3)\}$$

Now calculate RHS. First find A' and B' using $A'(x) = 1 - A(x)$ and $B'(x) = 1 - B(x)$:

$$A' = \{(a, 1 - 0.5), (b, 1 - 0.8), (c, 1 - 0.0), (d, 1 - 0.3)\} = \{(a, 0.5), (b, 0.2), (c, 1.0), (d, 0.7)\}$$

$$B' = \{(a, 1 - 0.2), (b, 1 - 1.0), (c, 1 - 0.1), (d, 1 - 0.7)\} = \{(a, 0.8), (b, 0.0), (c, 0.9), (d, 0.3)\}$$

Calculate $A' \cap B'(x) = \min[A'(x), B'(x)]$.

- $x = a: \min(0.5, 0.8) = 0.5$
- $x = b: \min(0.2, 0.0) = 0.0$
- $x = c: \min(1.0, 0.9) = 0.9$
- $x = d: \min(0.7, 0.3) = 0.3$

$$\text{RHS: } A' \cap B' = \{(a, 0.5), (b, 0.0), (c, 0.9), (d, 0.3)\}$$

Since LHS = RHS, the property $(A \cup B)' = A' \cap B'$ is shown to be true.

(b) Set Algebra Proofs

[7 Marks]

Problem: For the two sets A and B prove that:

(i) $A - (B \cup C) = (A - B) \cap (A - C)$

(ii) $A - (B \cap C) = (A - B) \cup (A - C)$

We use the definition of set difference: $x \in X - Y \iff x \in X$ and $x \notin Y$, and logical equivalences.

Proof of (i) $A - (B \cup C) = (A - B) \cap (A - C)$:

Let x be an arbitrary element.

$$\begin{aligned} x \in A - (B \cup C) &\iff x \in A \text{ and } x \notin (B \cup C) \\ &\iff x \in A \text{ and } \neg(x \in B \text{ or } x \in C) \\ &\iff x \in A \text{ and } (x \notin B \text{ and } x \notin C) \quad [\text{By De Morgan's Law}] \\ &\iff (x \in A \text{ and } x \notin B) \text{ and } (x \in A \text{ and } x \notin C) \quad [\text{Distributive}] \\ &\iff x \in (A - B) \text{ and } x \in (A - C) \\ &\iff x \in (A - B) \cap (A - C) \end{aligned}$$

Since this holds for any x , the sets are equal.

Proof of (ii) $A - (B \cap C) = (A - B) \cup (A - C)$:

Let x be an arbitrary element.

$$\begin{aligned} x \in A - (B \cap C) &\iff x \in A \text{ and } x \notin (B \cap C) \\ &\iff x \in A \text{ and } \neg(x \in B \text{ and } x \in C) \\ &\iff x \in A \text{ and } (x \notin B \text{ or } x \notin C) \quad [\text{By De Morgan's Law}] \\ &\iff (x \in A \text{ and } x \notin B) \text{ or } (x \in A \text{ and } x \notin C) \quad [\text{Distributive}] \\ &\iff x \in (A - B) \text{ or } x \in (A - C) \\ &\iff x \in (A - B) \cup (A - C) \end{aligned}$$

Since this holds for any x , the sets are equal.

Question 4

14 Marks

(a) Mathematical Induction

[7 Marks]

Problem: By the principle of mathematical induction, show that

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}, \quad n \geq 1$$

Let the given statement be $P(n) : 1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$

Step 1: Base Case

For $n = 1$, we need to prove $P(1)$ is true.

$$\text{LHS} = 1^2 = 1$$

$$\text{RHS} = \frac{1(1+1)(2(1)+1)}{6} = \frac{1(2)(3)}{6} = \frac{6}{6} = 1$$

Since $\text{LHS} = \text{RHS}$, $P(1)$ is true.

Step 2: Inductive Hypothesis

Assume that the statement $P(k)$ is true for some arbitrary positive integer k . That is:

$$1^2 + 2^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6} \quad \text{--- (Equation 1)}$$

Step 3: Inductive Step

We must prove that $P(k+1)$ is true. We need to show:

$$1^2 + 2^2 + \dots + k^2 + (k+1)^2 = \frac{(k+1)((k+1)+1)(2(k+1)+1)}{6} = \frac{(k+1)(k+2)(2k+3)}{6}$$



Starting with the LHS of $P(k+1)$:

$$\begin{aligned}
\text{LHS} &= [1^2 + 2^2 + \dots + k^2] + (k+1)^2 \\
&= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 \quad [\text{Substituting from Equation 1}] \\
&= \frac{k(k+1)(2k+1) + 6(k+1)^2}{6} \\
&= \frac{(k+1)[k(2k+1) + 6(k+1)]}{6} \quad [\text{Taking (k+1) common}] \\
&= \frac{(k+1)[2k^2 + k + 6k + 6]}{6} \\
&= \frac{(k+1)[2k^2 + 7k + 6]}{6} \\
&= \frac{(k+1)[2k^2 + 4k + 3k + 6]}{6} \quad [\text{Factorizing the quadratic}] \\
&= \frac{(k+1)[2k(k+2) + 3(k+2)]}{6} \\
&= \frac{(k+1)(k+2)(2k+3)}{6}
\end{aligned}$$

This matches the required RHS. Thus, $P(k+1)$ is true whenever $P(k)$ is true.

Conclusion: By the Principle of Mathematical Induction, the statement holds true for all $n \geq 1$.

(b) Fuzzy Relations Matrices

[7 Marks]

Problem: If R and S are fuzzy relations given by matrices:

$$R = \begin{bmatrix} 0.5 & 0.1 \\ 0.2 & 0.9 \\ 0.8 & 0.6 \end{bmatrix}, \quad S = \begin{bmatrix} 0.6 & 0.5 \\ 0.4 & 0.8 \\ 0.7 & 0.9 \end{bmatrix}$$

Then find $R \cup S, R \cap S, \overline{R}, \overline{S}$.

1. Fuzzy Union $R \cup S$:

The element-wise maximum of the two matrices: $(r \cup s)_{ij} = \max(r_{ij}, s_{ij})$

$$R \cup S = \begin{bmatrix} \max(0.5, 0.6) & \max(0.1, 0.5) \\ \max(0.2, 0.4) & \max(0.9, 0.8) \\ \max(0.8, 0.7) & \max(0.6, 0.9) \end{bmatrix} = \begin{bmatrix} \mathbf{0.6} & \mathbf{0.5} \\ \mathbf{0.4} & \mathbf{0.9} \\ \mathbf{0.8} & \mathbf{0.9} \end{bmatrix}$$

2. Fuzzy Intersection $R \cap S$:

The element-wise minimum of the two matrices: $(r \cap s)_{ij} = \min(r_{ij}, s_{ij})$

$$R \cap S = \begin{bmatrix} \min(0.5, 0.6) & \min(0.1, 0.5) \\ \min(0.2, 0.4) & \min(0.9, 0.8) \\ \min(0.8, 0.7) & \min(0.6, 0.9) \end{bmatrix} = \begin{bmatrix} \mathbf{0.5} & \mathbf{0.1} \\ \mathbf{0.2} & \mathbf{0.8} \\ \mathbf{0.7} & \mathbf{0.6} \end{bmatrix}$$

3. Fuzzy Complement $\overline{R}$:

The element-wise complement from 1: $(\overline{r})_{ij} = 1 - r_{ij}$

$$\overline{R} = \begin{bmatrix} 1 - 0.5 & 1 - 0.1 \\ 1 - 0.2 & 1 - 0.9 \\ 1 - 0.8 & 1 - 0.6 \end{bmatrix} = \begin{bmatrix} \mathbf{0.5} & \mathbf{0.9} \\ \mathbf{0.8} & \mathbf{0.1} \\ \mathbf{0.2} & \mathbf{0.4} \end{bmatrix}$$

4. Fuzzy Complement $\overline{S}$:

The element-wise complement from 1: $(\overline{s})_{ij} = 1 - s_{ij}$

$$\overline{S} = \begin{bmatrix} 1 - 0.6 & 1 - 0.5 \\ 1 - 0.4 & 1 - 0.8 \\ 1 - 0.7 & 1 - 0.9 \end{bmatrix} = \begin{bmatrix} \mathbf{0.4} & \mathbf{0.5} \\ \mathbf{0.6} & \mathbf{0.2} \\ \mathbf{0.3} & \mathbf{0.1} \end{bmatrix}$$